Literature Survey and Architecture Decision

Industrial Recommender Data-Generating Processes / 工业推荐 DGP

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Abstract

This review asks what simulator architecture can credibly test industrial short-video Feed and Local-service recommendation without leaking a hand-authored scoring formula to the evaluated rankers. We screened 1,500 OpenAlex candidates, attempted five Semantic Scholar expansion queries, retained 64 papers, and audited the repository's V1–V3 behavior kernels. V3 fixes correctness and calibration defects but remains a feature-derived formula world. The selected research direction is a partially observed neural structural causal world-model ensemble, with explicit slate competition, multi-action sequence generation, session and provider dynamics, paired exogenous noise, and randomized-log falsification. Public data is useful but not mandatory; without any real or randomized evidence, the result remains a synthetic stress world rather than a validated digital twin.

1 Research question and scope / 研究问题

The operational question is not whether a simulator can generate plausible-looking clicks. It is whether it can distinguish recall failure, coarse-rank loss, fine-rank error, mix displacement, calibration error, and long-horizon behavior under policy intervention. The target world contains play start, censored watch time, slide, engagement, negative feedback, POI and transaction funnels, session exit and return, supply response, multiple queues, and policy-induced distribution shift.

核心问题不是"能不能生成像真的 click"，而是能否在策略干预后，区分召回、粗排、精排、混排、校准和长期行为的责任，并对不确定性保持诚实。

2 Search methodology / 检索方法

OpenAlex broad, recent, methodology, and exact-seed searches returned 1,500 candidate records. Five Semantic Scholar Graph API queries were attempted with bounded retry and 1.1-second spacing; all returned HTTP 429 and no API key was available. This is recorded as a source limitation. DOI and normalized-title deduplication plus domain filtering produced 64 papers from 2010–2026.

Methodology group	Papers
Causal and off-policy evaluation	26
Ecosystem and feedback-loop simulation	12
LLM and generative-agent simulators	6
Configurable and mechanistic simulators	7
Supporting recommendation methodology	4
Sequential and long-horizon recommendation	3
Learned and data-driven simulators	6

Citation count was used for discovery, not as a quality score. Exact seeds preserve recent or low-citation systems including RecSim NG, KuaiSim, SARDINE, T-RECS, Virtual-Taobao, RL4RS, and Sim2Rec.

3 Audit of the current V1–V3 world / 当前系统审计

V1 generates immediate responses with mostly additive logits. V2 adds heavy-tailed population draws and a nonlinear stay adjustment derived from fields also visible to the rankers. V3 repairs correlated random streams, changes long-view semantics, and aligns several aggregate behavior rates to KuaiRand, but it retains a hand-authored response kernel.

The current long-view ladder therefore solves an artificially favorable tabular problem. XGBoost's pointwise AUC is 0.5953 versus logistic regression at 0.5934. Yet request-level audit regret is worse: 0.0922 versus 0.0393. The hand-authored rule reaches 0.0160. V3 has not shown that boosted trees produce better slates or long-horizon behavior.

V3 对齐的是总体 play、like、negative 和 stay 等边际量，没有对齐条件分布、联合行为、session transition、policy effect 或 intervention recovery。因此它是 calibration epoch，不是行为建模范式升级。

4 Literature landscape / 文献结论

4.1 Configurable and mechanistic simulators

RecSim makes latent preferences, user-state dynamics, familiarity, choice, and response configurable. RecSim NG adds probabilistic multi-agent modeling, latent-variable learning, and accelerator execution. SARDINE represents recommendation-induced user changes and feedback from biased logged data. T-RECS makes ecosystem-level effects observable. These systems provide the right causal scaffold but do not guarantee empirical realism.

4.2 Learned and data-driven simulators

Virtual-Taobao learns a virtual retail environment and explicitly considers simulator exploitation. KuaiSim covers list-wise response, multi-behavior feedback, sessions, and cross-session retention. RL4RS separates simulator evaluation, policy evaluation in simulation, and counterfactual evaluation. Sim2Rec uses a set of possible environments to reduce over-trust in one learned simulator. The consensus is that one-step predictive accuracy is insufficient because errors compound and policies seek unsupported regions.

4.3 Causal and slate evaluation

IPS, SNIPS, counterfactual risk minimization, doubly robust estimators, and structured slate estimators address bias and variance under specific assumptions. They do not solve absent action support. A simulator and OPE have complementary roles: the simulator rolls out mechanisms; randomized/exploratory logs constrain intervention effects and expose extrapolation.

4.4 LLM and generative agents

LLM agents add semantic profiles, memory, emotion, rationales, and flexible actions. They are promising for conversational recommendation, semantic intent, and rare scenario generation. They are not a suitable primary engine for tens of millions of reproducible numeric Feed events because inference is expensive, calibration is difficult, paired counterfactual noise is awkward, and model upgrades silently change the environment.

5 Architecture comparison / 架构比较

Alternative	Strength	Limitation and decision
More nonlinear formulas	Fast, transparent, exact truth	Preserves feature leakage and designer bias; retain only as deterministic test worlds.
One-step learned response	Easy calibration and training	Misses trajectory, slate competition, and compounding error; use only as one component.
Pure autoregressive model	Learns joint temporal patterns	Does not identify interventions outside logging support; place inside a causal scaffold.
Pure LLM agents	Rich semantics and rare scenarios	Slow, unstable, hard to calibrate; use as scenario generator and challenger.
Pure agent-based model	Explicit interference and provider behavior	Parameters remain hand-authored; use as structural scaffold.
Hybrid neural SCM ensemble	Learned distributions, explicit interventions, uncertainty	Highest validation cost but the only candidate satisfying the target; selected.

6 Selected architecture / 选定方向

The selected design contains six independent authorities:

1. A learned correlated population and catalog process for users, creators, items, lifecycle, supply arrival, and inventory.
2. A latent state-space model for stable preferences, multi-interest, novelty, fatigue, satisfaction, commercial intent, and session intent. Serving sees noisy projections only.
3. A set/attention slate encoder and autoregressive multi-action decoder for play, censored stay, slide, engagement, negative feedback, POI, and transaction events.
4. Recurrent/Transformer state transitions plus survival heads for session exit, return delay, active day, and provider response.
5. Versioned exogenous noise and an ensemble/posterior over plausible worlds for paired potential outcomes and epistemic uncertainty.
6. A validation authority based on standard logs, randomized/exploratory traffic, held-out policies, historical A/Bs, and synthetic stress worlds.

The policy must never access latent simulator state. Long-term value is measured from emergent trajectories and externally accepted exchange rates; it is not injected as a direct reward formula into behavior generation.

7 Data requirement / 数据要求

Public data is optional. Proprietary randomized traffic with propensities is preferred, followed by proprietary observational logs plus historical orthogonal A/Bs. Public randomized logs provide reproducible external falsification. Public observational logs test shape and pipelines. Fully synthetic worlds test mechanisms only. Without real or randomized evidence, no architecture can honestly claim real-user fidelity.

8 Acceptance and falsification / 验收与证伪

V4 must pass five gates: joint/conditional distribution calibration; free-running sequence calibration; held-out intervention recovery; frozen-policy ordering agreement; and anti-exploitation across an ensemble of plausible worlds. Required negative controls include shuffled treatment, future-feature leakage, duplicated events, missing propensity, and an intentionally reward-hacking policy.

Every synthetic launch must report empirical support distance, ensemble disagreement, rollout drift, propensity coverage, and whether the tested policy remains inside the calibrated region. A zero-width synthetic confidence interval is not evidence if world-model uncertainty is omitted.

9 Research gaps and final decision / 研究缺口与结论

No public system jointly validates short-video multi-action response, censored watch time, session return, provider/supply feedback, Local transactions, and policy interventions. Marginal calibration is common; intervention recovery and policy-order agreement are rare. LLM agents improve semantics but not calibrated numeric throughput.

The implementation decision is therefore intentionally conditional: preserve V1–V3 as historical test epochs; prototype the hybrid neural SCM on a frozen request-level dataset; compare it with V3 on the five falsification gates; promote it only after held-out intervention recovery and policy-order agreement. Do not create a V4 by adding another formula.

A Full paper catalog

Year	Title	Methodology group	Cites
2026	The urban impact of AI: modelling feedback loops in location-based recommender systems	Ecosystem and feedback-loop simulation	3
2026	Causal Recommendation under Confounder Observability: A Short Survey	Causal and off-policy evaluation	0
2025	LLM-Powered User Simulator for Recommender System	LLM and generative-agent simulators	22
2025	A LLM-based Controllable, Scalable, Human-Involved User Simulator Framework for Conversational Recommender Systems	LLM and generative-agent simulators	10

Year	Title	Methodology group	Cites
2025	RecUserSim: A Realistic and Diverse User Simulator for Evaluating Conversational Recommender Systems	LLM and generative-agent simulators	8
2025	Breaking Feedback Loops in Recommender Systems with Causal Inference	Ecosystem and feedback-loop simulation	4
2025	LLM-Based User Simulation for Low-Knowledge Shilling Attacks on Recommender Systems	LLM and generative-agent simulators	1
2024	On Generative Agents in Recommendation	LLM and generative-agent simulators	97
2024	Attention Is Not the Only Choice: Counterfactual Reasoning for Path-Based Explainable Recommendation	Causal and off-policy evaluation	45
2024	Causally estimating the effect of YouTube's recommender system using counterfactual bots	Causal and off-policy evaluation	39
2024	Reinforced Path Reasoning for Counterfactual Explainable Recommendation	Causal and off-policy evaluation	31
2024	Counterfactual Explanation for Fairness in Recommendation	Causal and off-policy evaluation	28
2024	GNNUERS: Fairness Explanation in GNNs for Recommendation via Counterfactual Reasoning	Causal and off-policy evaluation	23
2024	Oh, Behave! Country Representation Dynamics Created by Feedback Loops in Music Recommender Systems	Ecosystem and feedback-loop simulation	11
2024	Off-Policy Evaluation of Slate Bandit Policies via Optimizing Abstraction	Causal and off-policy evaluation	7
2024	SARDINE: Simulator for Automated Recommendation in Dynamic and Interactive Environments	Configurable and mechanistic simulators	4
2024	A LLM-based Controllable, Scalable, Human-Involved User Simulator Framework for Conversational Recommender Systems	LLM and generative-agent simulators	2
2024	Distributional Off-Policy Evaluation for Slate Recommendations	Causal and off-policy evaluation	2
2023	CIRS: Bursting Filter Bubbles by Counterfactual Interactive Recommender System	Causal and off-policy evaluation	89
2023	UserSimCRS: A User Simulation Toolkit for Evaluating Conversational Recommender Systems	Supporting recommendation methodology	25
2023	Generative Slate Recommendation with Reinforcement Learning	Sequential and long-horizon recommendation	21
2023	RL4RS: A Real-World Dataset for Reinforcement Learning based Recommender System	Learned and data-driven simulators	15
2023	Sim2Rec: A Simulator-based Decision-making Approach to Optimize Real-World Long-term User Engagement in Sequential Recommender Systems	Learned and data-driven simulators	11
2023	KuaiSim: A Comprehensive Simulator for Recommender Systems	Learned and data-driven simulators	8
2023	A Complete Framework for Offline and Counterfactual Evaluations of Interactive Recommendation Systems	Causal and off-policy evaluation	3
2023	Navigating the Feedback Loop in Recommender Systems: Insights and Strategies from Industry Practice	Ecosystem and feedback-loop simulation	3
2023	SARDINE: A Simulator for Automated Recommendation in Dynamic and Interactive Environments	Configurable and mechanistic simulators	1
2022	Disentangling Long and Short-Term Interests for Recommendation	Causal and off-policy evaluation	119
2022	Breaking Feedback Loops in Recommender Systems with Causal Inference	Ecosystem and feedback-loop simulation	11
2022	Enhancing Counterfactual Evaluation and Learning for Recommendation Systems	Causal and off-policy evaluation	1
2021	Break the Loop: Gender Imbalance in Music Recommenders	Ecosystem and feedback-loop simulation	89
2021	Counterfactual Learning and Evaluation for Recommender Systems: Foundations, Implementations, and Recent Advances	Causal and off-policy evaluation	51
2021	RL4RS: A Real-World Benchmark for Reinforcement Learning based Recommender System.	Learned and data-driven simulators	13
2021	RecSim NG: Toward Principled Uncertainty Modeling for Recommender Ecosystems	Configurable and mechanistic simulators	9
2021	T-RECS: A Simulation Tool to Study the Societal Impact of Recommender Systems	Configurable and mechanistic simulators	3
2021	RL4RS: A Real-World Dataset for Reinforcement Learning based Recommender System	Learned and data-driven simulators	1
2020	SSE-PT: Sequential Recommendation Via Personalized Transformer	Supporting recommendation methodology	258
2020	Self-Supervised Reinforcement Learning for Recommender Systems	Causal and off-policy evaluation	217
2020	Evaluating Conversational Recommender Systems via User Simulation	Configurable and mechanistic simulators	84
2020	Bias in Search and Recommender Systems	Ecosystem and feedback-loop simulation	76
2020	Counterfactual Evaluation of Slate Recommendations with Sequential Reward Interactions	Causal and off-policy evaluation	47
2020	Information Theoretic Counterfactual Learning from Missing-Not-At-Random Feedback	Causal and off-policy evaluation	28
2020	Feedback Loop and Bias Amplification in Recommender Systems	Ecosystem and feedback-loop simulation	21
2020	Do Offline Metrics Predict Online Performance in Recommender Systems?	Supporting recommendation methodology	11
2019	Top-K Off-Policy Correction for a REINFORCE Recommender System	Causal and off-policy evaluation	400
2019	Reinforcement Learning to Optimize Long-term User Engagement in Recommender Systems	Causal and off-policy evaluation	245
2019	Degenerate Feedback Loops in Recommender Systems	Ecosystem and feedback-loop simulation	151
2019	Virtual-Taobao: Virtualizing Real-World Online Retail Environment for Reinforcement Learning	Learned and data-driven simulators	147

Year	Title	Methodology group	Cites
2019	Offline Evaluation to Make Decisions About PlaylistRecommendation Algorithms	Causal and off-policy evaluation	142
2019	SlateQ: A Tractable Decomposition for Reinforcement Learning with Recommendation Sets	Sequential and long-horizon recommendation	116
2019	Debiasing the Human-Recommender System Feedback Loop in Collaborative Filtering	Ecosystem and feedback-loop simulation	80
2019	RecSim: A Configurable Simulation Platform for Recommender Systems	Configurable and mechanistic simulators	50
2018	How algorithmic confounding in recommendation systems increases homogeneity and decreases utility	Ecosystem and feedback-loop simulation	325
2018	Offline A/B Testing for Recommender Systems	Causal and off-policy evaluation	135
2018	RecoGym: A Reinforcement Learning Environment for the problem of Product Recommendation in Online Advertising	Configurable and mechanistic simulators	60
2017	Deconvolving Feedback Loops in Recommender Systems	Ecosystem and feedback-loop simulation	43
2016	Fusing Similarity Models with Markov Chains for Sparse Sequential Recommendation	Sequential and long-horizon recommendation	736
2016	Off-policy evaluation for slate recommendation	Causal and off-policy evaluation	39
2015	The self-normalized estimator for counterfactual learning	Causal and off-policy evaluation	202
2015	Batch learning from logged bandit feedback through counterfactual risk minimization	Causal and off-policy evaluation	189
2015	Counterfactual Risk Minimization: Learning from Logged Bandit Feedback	Causal and off-policy evaluation	127
2015	Doubly Robust Off-policy Value Evaluation for Reinforcement Learning	Causal and off-policy evaluation	81
2015	Counterfactual Risk Minimization	Causal and off-policy evaluation	70
2010	A contextual-bandit approach to personalized news article recommendation	Supporting recommendation methodology	2579

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